A home valuation without its asking price

My approach: learn nonlinear price relationships from allowed property information

PROPERTY & LOT

Size, rooms, parking and valid lot measurements

SYNTHETIC LOCATION

Property type, style, municipality and community

TIME & DESCRIPTION

Listing date and fixed condition / amenity cues

Histogram gradient boosting

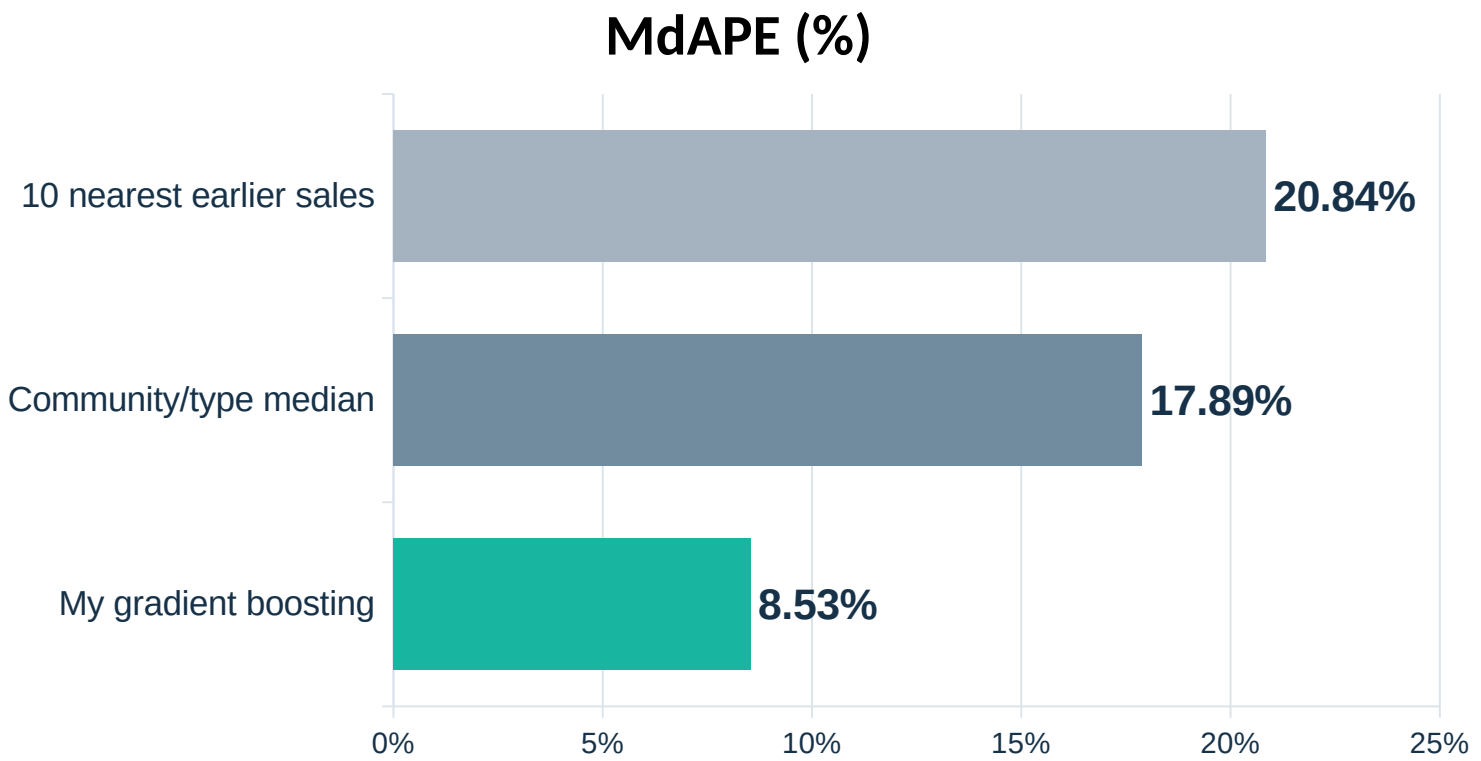
Learn log(sale price)

Unknown measurements stay missing
Category encoding fits within each training fold

Excluded: list price • post-sale fields • AVMs • IDs

Median error fell to 8.53%

Comparison on the same 10,069 held-out homes and original sale-price labels



8.53%
Median absolute % error

11.00%
Mean absolute % error

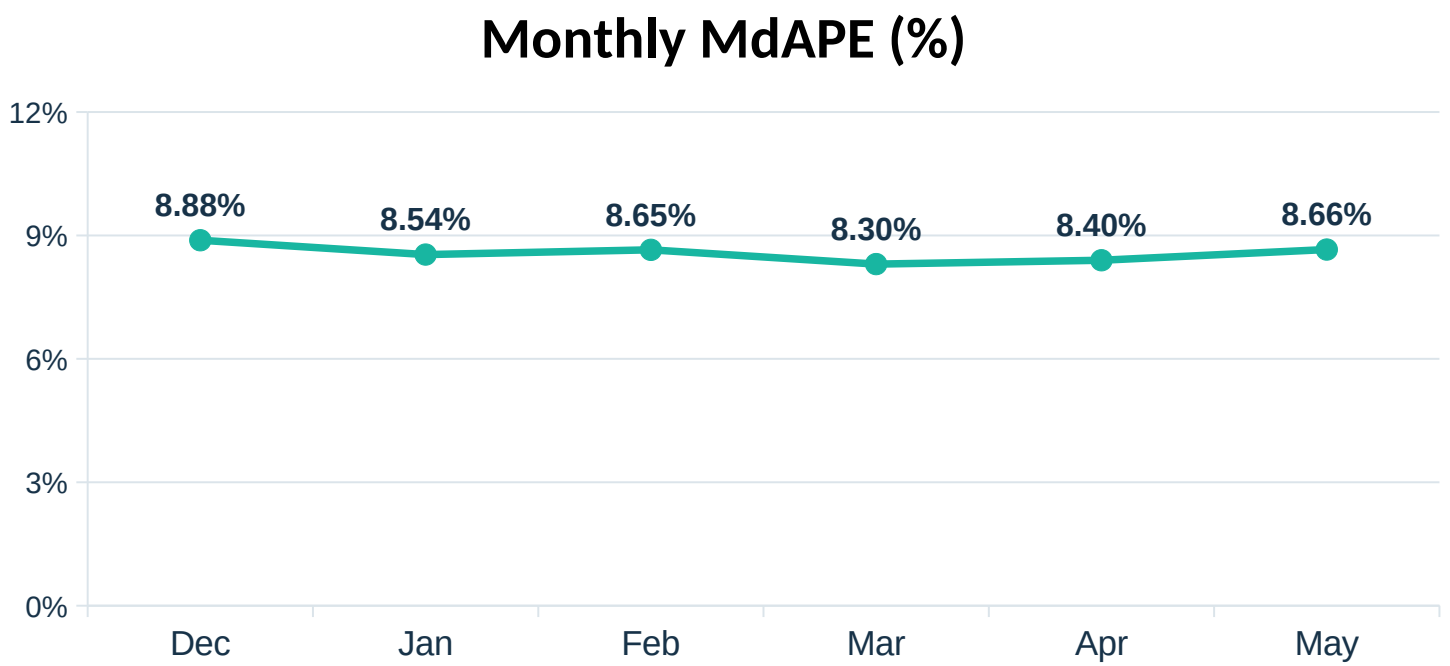
10,069
Held-out homes

Lower error is better. Reference baselines: retained original-data run; identical homes and labels verified.

Performance over time. Checked final output.

Each validation month uses only sales before that month · December 2025–May 2026

Monthly median absolute percentage error



3,974

off-market predictions

Final fit: all 150,000 original training sales

- ✓ Every test ID exactly once
- ✓ Positive, finite prices
- ✓ Official CSV checker passed

submission_off_market.csv

Validation is not the hidden test. Future market conditions remain a limitation.